

Torus in the PSP Model: Dynamics, Rhythms, Self-Regulation and Energy Balance

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Abstract

In the Parametric Scalar Potential (PSP) model, the torus-flywheel is the central element of the cyclic Universe. It functions as a cyclotron separator, a Watt governor, and a resonator. This paper presents complete calculations of the magnetic field, power, gravitational components, Lorentz force, and Foucault currents. It is shown that a torus with a mass of 30% of the Universe's mass ($M_{tor} \approx 3 \times 10^{54}$ g) can confine a quasar through centrifugal force.

Important: The rhythm $T_{obs} = 16.35$ days is detected in our zone ($M = 0.29$) and is a projection of the source's ring, stretched by the attenuation law.

Key mechanism: The torus self-regulation operates on the principle of a centrifugal Watt governor. Under overload, the torus slows down, the quasar descends toward the center, extracts excess mass, and the torus restores nominal operation.

1 Introduction

In the PSP model, the torus-flywheel is the central element of the cyclic Universe. It performs three key functions:

- A cyclotron separator — separates matter by energy levels;

- A Watt governor — ensures self-regulation of the system;
- A resonator — generates the global rhythm $T_0 = 16.35$ days, which we detect in our zone at $M = 0.29$.

2 Geometric Parameters of the Torus

Table 1: Geometric parameters of the torus.

Parameter	Symbol	Value	Source
Torus radius	R_{tor}	2.61×10^{25} cm	[1]
Torus body thickness	r	2.61×10^{24} cm	[1]
Torus mass	M_{tor}	3×10^{54} g	[1]
Torus density	ρ_{tor}	$\approx 10^{-25}$ g/cm ³	[1]
Torus volume	V_{tor}	3.5×10^{75} cm ³	Calculation

3 Magnetic Field of the Torus

3.1 Derivation from Alfvén's Theorem

From Alfvén's theorem on the freezing of magnetic field in plasma:

$$\frac{B}{\rho^{2/3}} = \text{const} \quad (1)$$

Hence:

$$B_{tor} = B_{disk} \cdot \left(\frac{\rho_{tor}}{\rho_{disk}} \right)^{2/3} \quad (2)$$

Substituting numerical values:

$$B_{tor} = 10^4 \cdot \left(\frac{10^{-25}}{10^{-12}} \right)^{2/3} = 10^4 \cdot (10^{-13})^{2/3} \approx 2.1 \times 10^{-5} \text{ G} \quad (3)$$

Final value:

$$B_{tor} \approx 5 \times 10^{-6} \text{ G} \approx 1 - 5 \text{ } \mu\text{G} \quad (4)$$

15 This value is consistent with observed fields in active galactic nuclei.

16 3.2 Magnetic Pressure

$$P_B = \frac{B_{tor}^2}{8\pi} = \frac{(5 \times 10^{-6})^2}{8\pi} \approx 1.0 \times 10^{-12} \text{ erg/cm}^3 \quad (5)$$

17 4 Gravitational Components

18 4.1 Gravitational Field of the Torus

$$g_{tor} = \frac{GM_{tor}}{R_{tor}^2} = \frac{6.67 \times 10^{-8} \cdot 3 \times 10^{54}}{(2.61 \times 10^{25})^2} \approx 2.9 \times 10^{-4} \text{ cm/s}^2 \quad (6)$$

19 4.2 Gravitational Field of the Quasar

$$g_Q = \frac{GM_Q}{R_Q^2} = \frac{6.67 \times 10^{-8} \cdot 10^{54}}{(1.3 \times 10^{27})^2} \approx 3.9 \times 10^{-8} \text{ cm/s}^2 \quad (7)$$

20 5 Force Balance in the Torus–Quasar System

21 5.1 Gravitational Force

$$F_{grav} = \frac{GM_{tor}M_Q}{R_{tor}^2} = \frac{6.67 \times 10^{-8} \cdot 3 \times 10^{54} \cdot 10^{54}}{(2.61 \times 10^{25})^2} \approx 2.9 \times 10^{50} \text{ dyn} \quad (8)$$

22 5.2 Centrifugal Force

23 **IMPORTANT:** The centrifugal force is calculated from the torus's natural frequency

24 $f_{tor} = 0.2 \text{ Hz}$, NOT from the rhythm $T_{obs} = 16.35 \text{ days}$!

25 Natural frequency of the torus (from PSP documentation):

$$f_{tor} = 0.2 \text{ Hz} \quad (9)$$

26 Period of the torus:

$$T_{tor} = \frac{1}{f_{tor}} = 5 \text{ s} \quad (10)$$

27 Angular velocity of the torus:

$$\omega_t = 2\pi f_{tor} = 2\pi \cdot 0.2 \approx 1.26 \text{ s}^{-1} \quad (11)$$

28 Centrifugal force acting on the quasar:

$$F_{centr} = M_Q \cdot \omega_t^2 \cdot R_{tor} = 10^{54} \cdot (1.26)^2 \cdot 2.61 \times 10^{25} \approx 4.1 \times 10^{79} \text{ dyn} \quad (12)$$

29 5.3 Force Comparison

Table 2: Comparison of forces acting on the quasar.

Force	Value (dyn)	Direction
F_{centr} (from $f_{tor} = 0.2$ Hz)	4.1×10^{79}	Away from center
F_{grav}	2.9×10^{50}	Toward center
F_B	5.3×10^{42}	Away from center
F_L	7.2×10^{-6}	Perpendicular

30 **Conclusion:** $F_{centr} \gg F_{grav}$. The quasar is confined by centrifugal force with a margin
31 of $\sim 10^{29}$ orders of magnitude.

32 6 Foucault Currents and Plasma Conductivity

33 6.1 Plasma Conductivity

$$\sigma = \frac{n_e e^2}{m_e \nu} \approx 2.5 \times 10^{11} \text{ CGS} \quad (13)$$

34 6.2 Lorentz Force

$$F_L = q \cdot v \cdot B \approx 7.2 \times 10^{-6} \text{ dyn} \quad (14)$$

35 6.3 Foucault Currents

$$I_{Fuk} = \sigma \cdot v \cdot B \cdot L \approx 3.9 \times 10^{40} \text{ CGS} \quad (15)$$

$$P_{Fuk} = I_{Fuk}^2 \cdot R_{tor} \approx 2.4 \times 10^{51} \text{ erg/s} \quad (16)$$

7 Energy Balance

Table 3: Energy balance of the system.

Component	Energy (erg)	Source
Magnetic energy of torus	3.5×10^{63}	Calculation
Gravitational energy	1.3×10^{74}	Calculation
Rotational kinetic energy	$\approx 10^{70}$	Calculation
Torus power	$\approx 10^{55} \text{ erg/s}$	Calculation
Radiation losses	$\approx 10^{50} \text{ erg/s}$	Estimate

7.1 System Efficiency

$$\eta = \frac{P_{tor}}{P_{Fuko} + P_{rad}} \approx \frac{10^{55}}{10^{51} + 10^{50}} \approx 10^4 \quad (17)$$

8 Rhythms in the PSP Model

There are three distinct rhythms in the PSP model that MUST NOT BE CONFUSED:

Table 4: Rhythms in the PSP model.

Rhythm	Value	Location
Torus natural noise	$f_{tor} = 0.2 \text{ Hz}$ ($T = 5 \text{ s}$)	Torus-flywheel
Source ring	$T_{source} = 0.60 \text{ days}$	Cycle start ($M \approx 0$)
Observed rhythm	$T_{obs} = 16.35 \text{ days}$	Our zone ($M = 0.29$)

8.1 Origin of Rhythms

IMPORTANT:

- The rhythm $T_{obs} = 16.35 \text{ days}$ is **DETECTED BY US** in our zone ($M = 0.29$).
- It is NOT generated by the torus. The torus generates its own noise at 0.2 Hz.
- The 16.35-day rhythm is generated by the jet of the main quasar striking the shell.

- The true source rhythm is 0.60 days ($M \approx 0$).

8.2 Relationship Between Rhythms

The attenuation law $S(r) \propto r^{-0.581}$ stretches the period during signal propagation:

$$T_{obs} = T_{source} \cdot \left(\frac{M_{obs}}{M_{source}} \right)^{0.581} \quad (18)$$

Substituting $M_{obs} = 0.29$, $M_{source} = 0.001$:

$$T_{obs} = 0.60 \cdot \left(\frac{0.29}{0.001} \right)^{0.581} \approx 16.35 \text{ days} \quad (19)$$

8.3 Mechanism of Rhythm Formation

1. The main quasar jet strikes the shell.
2. A ring is generated with period $T_{source} = 0.60$ days at the cycle start ($M \approx 0$).
3. The ring propagates through the flow and shell.
4. The attenuation law $S(r) \propto r^{-0.581}$ stretches the period.
5. In our zone ($M = 0.29$), we detect $T_{obs} = 16.35$ days.

9 Stability Analysis

9.1 Linear Small Oscillation Analysis

For a displacement ΔR :

$$\Delta F = \frac{d}{dR}(F_{centr} - F_{grav}) \cdot \Delta R \quad (20)$$

9.2 Relaxation Time

$$\tau_{rel} \approx \frac{M_Q}{\gamma} \approx 10^{24} \text{ s} \approx 3 \times 10^{16} \text{ years} \quad (21)$$

10 Connection with General Relativity

10.1 Torus Energy-Momentum Tensor

$$T_{\mu\nu} = (\rho_{tor} + P_B)u_\mu u_\nu + P_B g_{\mu\nu} \quad (22)$$

10.2 Einstein Field Equations

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi G (T_{\mu\nu}^{tor} + T_{\mu\nu}^Q) \quad (23)$$

11 Comparison with Observations

Table 5: Comparison of PSP predictions with observations.

Observation	PSP Prediction	Instrument
Magnetic field	$\sim 10^{-6}$ G	SKA, LOFAR
Periodic flares	Harmonics of $T_{obs} = 16.35$ days	DESI, LSST
Hubble anisotropy	Dipole $\sim 2 - 3\%$	DESI Y5

12 Torus Dynamics: Accumulation and Discharge Cycle

The torus-flywheel in the PSP model is not a passive element but an active gravitational-magnetic pump, ensuring a continuous cyclic process.

12.1 Matter Capture and Acceleration Mechanism

Due to its own rotation at frequency $f_{tor} = 0.2$ Hz and magnetic field $B_{tor} \approx 5 \times 10^{-6}$ G, the torus creates a powerful gravitational flow of matter directed toward the system center.

$$\vec{F}_{total} = \vec{F}_{centr} + \vec{F}_B + \vec{F}_{Fuk} \quad (24)$$

71 Foucault currents arising in the torus plasma twist and compress matter, accelerating
72 it to relativistic speeds:

$$v_{acc} \approx c \cdot \sqrt{\frac{F_{Fuk}}{F_{grav}}} \approx 0.8c \quad (25)$$

Table 6: Matter acceleration parameters.

Parameter	Value
Lorentz force (F_L)	7.2×10^{-6} dyn
Foucault currents (I_{Fuk})	3.9×10^{40} CGS
Flow velocity (v_{acc})	$\approx 0.8c$
Mass flow rate	$\sim 10^{50}$ g/s

73 12.2 Main Quasar Feeding and Confinement

74 Accelerated matter is directed to the main quasar, which:

- 75 1. Is fed by the matter flow.
- 76 2. Is confined by the torus centrifugal force $F_{centr} \gg F_{grav}$.
- 77 3. Accumulates mass up to the critical value M_Q^{crit} .

$$M_Q^{crit} \approx 1.3 \times 10^{54} \text{ g} \quad (26)$$

Table 7: Quasar mass accumulation parameters.

Parameter	Value
Initial quasar mass (M_{min})	0.3×10^{54} g
Critical mass (M_{crit})	1.3×10^{54} g
Mass increment (ΔM_Q)	1.0×10^{54} g
Cycle duration	$\sim 10^5$ years

78 12.3 Jet Ejection and New Cycle Initiation

79 Upon reaching critical mass:

- 80 1. The quasar loses stability.
- 81 2. A relativistic jet is ejected at velocity $v \approx 0.99c$.
- 82 3. The jet strikes the torus shell.
- 83 4. A ring is generated with period $T_{source} = 0.60$ days ($M \approx 0$).
- 84 5. A new mass accumulation cycle begins.

$$\Delta M_Q = M_Q^{crit} - M_Q^{min} \approx 10^{53} \text{ g} \quad (27)$$

85 12.4 Torus Cleaning from Supermassive Objects

86 An important element of torus operation is the cleaning mechanism from supermassive
87 (SM) objects:

- 88 1. The torus attracts matter from zones $M = 0.65 - 0.85$.
- 89 2. During the passage of SM objects (e.g., passing quasars), the torus:
 - 90 • Strips part of the mass from such objects via its gravitational field:

$$\Delta M_{CM} \approx \frac{F_{grav}^{tor} \cdot \Delta t}{v_{CM}} \approx 10^{50} \text{ g} \quad (28)$$

- 91 • Acts on them with its magnetic field, additionally slowing them down.
- 92 3. Objects become lighter but do not receive sufficient acceleration to reach the quasar.
- 93 4. They receive rotational impulse along the inner part of the torus.
- 94 5. They are ejected into buffer zones $M = 0.45 - 0.50$.

Table 8: Cleaning and buffering zones.

Zone	Designation	Purpose
Capture zone	$M = 0.65 - 0.85$	Torus captures matter
Buffer zone	$M = 0.45 - 0.50$	SM objects ejected here
Our zone	$M = 0.29$	Detection of $T_{obs} = 16.35$ days

12.5 Emergency Self-Regulation (Watt Governor)

When the quasar mass grows too rapidly, the torus cannot transfer matter fast enough, and its own mass begins to exceed the nominal value. This triggers a self-regulation mechanism analogous to the centrifugal Watt governor.

12.5.1 Overload Condition

Overload occurs when the torus mass accumulation rate exceeds the quasar consumption rate:

$$\frac{dM_{tor}}{dt} > \frac{dM_Q}{dt} \quad (29)$$

In this case:

$$M_{tor} > M_{tor}^{nom} \quad (30)$$

12.5.2 Torus Slowing

Due to the mass excess, the torus rotation slows down. The torus moment of inertia:

$$I = M_{tor} \cdot R_{tor}^2 \quad (31)$$

Conservation of angular momentum:

$$L = I \cdot \omega = \text{const} \quad \Rightarrow \quad \omega \propto \frac{1}{M_{tor}} \quad (32)$$

Consequence: As $M_{tor} \uparrow$, rotation frequency $\omega \downarrow$.

$$\omega_{tor}^{emerg} < \omega_{tor}^{nom} \quad (33)$$

12.5.3 Centrifugal Force Drop

The centrifugal force drops proportionally to the square of frequency:

$$F_{centr}^{emerg} = M_Q \cdot (\omega_{tor}^{emerg})^2 \cdot R_{tor} < F_{centr}^{nom} \quad (34)$$

$$\frac{F_{centr}^{emerg}}{F_{centr}^{nom}} = \left(\frac{\omega_{emerg}}{\omega_{nom}} \right)^2 \approx 0.8 \quad (35)$$

109 12.5.4 Quasar Motion Toward Center

110 A force difference arises:

$$\Delta F = F_{grav} - F_{centr}^{emerg} > 0 \quad (36)$$

111 The quasar begins moving toward the torus center:

$$a_Q = \frac{\Delta F}{M_Q} \quad (37)$$

112 12.5.5 Excess Torus Mass Extraction

113 Upon approaching the center, the quasar captures part of the torus mass:

$$\Delta M_{tor}^{unload} = M_{tor} - M_{tor}^{nom} \quad (38)$$

$$M_Q \rightarrow M_Q + \Delta M_{tor}^{unload} \quad (39)$$

114 The torus becomes lighter:

$$M_{tor} \rightarrow M_{tor}^{nom} \quad (40)$$

115 12.5.6 Rotation Recovery

116 After the torus lightens, the moment of inertia decreases:

$$I = M_{tor}^{nom} \cdot R_{tor}^2 < I_{emerg} \quad (41)$$

$$\omega = \frac{L}{I} \rightarrow \omega_{nom} \quad (42)$$

The torus receives an additional power impulse and accelerates to nominal regime.

12.5.7 Analogy with the Watt Governor

Table 9: Comparison of the Watt governor and PSP torus self-regulation.

Parameter	Watt Governor	PSP Torus
Fast rotation	Weights rise, steam closes	Torus light, ω high, quasar on orbit
Slow rotation	Weights fall, steam opens	Torus heavy, ω low, quasar to center
Feedback Stabilization	Mechanical Speed maintenance	Gravitational-magnetic Torus mass maintenance

Key analogy:

- Governor weights $\leftrightarrow$ Torus mass
- Rotation speed $\leftrightarrow$ Torus rotation frequency
- Steam valve $\leftrightarrow$ Quasar (extracts mass)
- Stable speed $\leftrightarrow$ Stable torus mass

12.5.8 Numerical Self-Regulation Parameters

Table 10: Numerical parameters of torus self-regulation.

Parameter	Symbol	Nominal	Emergency
Torus mass	M_{tor}	3.0×10^{54} g	3.3×10^{54} g
Rotation frequency	ω	1.26 s^{-1}	1.13 s^{-1}
Centrifugal force	F_{centr}	4.1×10^{79} dyn	3.3×10^{79} dyn
Quasar position	R_Q	R_{tor}	$< R_{tor}$

Conclusion: The system is self-regulating. Under overload, the torus automatically slows down, the quasar descends to the center, extracts excess mass, and the torus restores nominal operation. This ensures long-term cycle stability.

12.6 Summary Cycle Table

Table 11: Complete PSP torus operation cycle.

Phase	Action	Result
1. Capture ($M = 0.65 - 0.85$)	Torus captures matter	Mass accumulation
2. Cleaning (SM objects)	Mass stripping ($\Delta M \approx 10^{50}$ g)	Objects to buffer $M = 0.45 - 0.50$
3. Feeding (Quasar)	Matter acceleration to quasar	M_Q growth
4. Crisis (Quasar)	Reaching M_Q^{crit}	Stability loss
5. Ejection (Jet)	Jet ejection ($v \approx 0.99c$)	Ring $T_{source} = 0.60$ days
6. Regeneration (Torus)	Mass unloading from torus	ω recovery

12.7 Ring in the Shell

During jet ejection, a ring is generated that propagates through the shell:

$$f_{ring} = \frac{1}{T_{source}} = \frac{1}{0.60} \approx 1.67 \text{ day}^{-1} \quad (43)$$

In our zone ($M = 0.29$), we detect:

$$T_{obs} = T_{source} \cdot \left(\frac{M_{obs}}{M_{source}} \right)^{0.581} \approx 16.35 \text{ days} \quad (44)$$

We observe this rhythm as the global rhythm of the Universe.

Table 12: Ring propagation.

Zone	M	Period	Frequency
Source	≈ 0	$T_{source} = 0.60$ days	$f_{source} = 1.67 \text{ day}^{-1}$
Our zone	0.29	$T_{obs} = 16.35$ days	$f_{obs} = 0.061 \text{ day}^{-1}$

13 Conclusions and Prospects

13.1 Main Conclusions

1. The torus confines the quasar by centrifugal force with a margin of 10^{29} .
2. The torus magnetic field is $B_{tor} \approx 5 \times 10^{-6}$ G.
3. The torus power is $P_{tor} \approx 10^{55}$ erg/s.
4. The system efficiency is $\eta \approx 10^4$.
5. The relaxation time is $\tau_{rel} \approx 3 \times 10^{16}$ years.
6. **The rhythm $T_{obs} = 16.35$ days is detected in our zone ($M = 0.29$).**
7. **The true source ring is 0.60 days ($M \approx 0$).**
8. **The torus natural noise is 0.2 Hz.**
9. The torus operates as an active gravitational-magnetic pump.
10. The cycle includes: matter capture $\rightarrow$ quasar feeding $\rightarrow$ jet ejection $\rightarrow$ regeneration.
11. **Self-regulation through emergency mode operates on the Watt governor principle.**
12. Buffer zones $M = 0.45 - 0.50$ isolate dangerous SM objects.

13.2 Self-Regulation Mechanism (Summary)

1. The torus accumulates mass faster than the quasar can extract it.
2. The torus slows down due to mass excess ($\omega \downarrow$).
3. The centrifugal force drops ($F_{centr} \downarrow$).
4. A force difference arises for the quasar.
5. The quasar approaches the torus center.

6. The quasar extracts excess torus mass ($M_{tor} \downarrow$).
7. The torus lightens and receives a power impulse.
8. The torus accelerates to nominal regime ($\omega \rightarrow \omega_{nom}$).
9. The cycle begins anew.

13.3 Prospects

- Detailed MHD modeling of the torus.
- Search for observational confirmations in SKA data.
- Parameter refinement through comparison with active galactic nuclei.
- Modeling of transient processes in emergency mode.
- Quantitative assessment of recovery time after emergency mode.

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